

Super-Liouville conformal blocks

Sphere four-point and torus two-point decompositions
in the local Type 0B amplitude convention

Machine note accompanying the standalone source package

Local source snapshot: 24 September 2026

Abstract

This note specifies the coefficients, nonchiral contractions and numerical block algorithms used by the local amplitude code. It covers four NS, two NS and two R, and four R external fields on the sphere, and NS or R external fields on the torus. The three-point constants are explicit. The sphere scattering half-sum Ramond dictionary is retained; its conversion to the separate equal-metric sewing convention is described, rather than silently applied. Fresh coefficient and sphere crossing checks accompany the package. The Type 0B sphere amplitude plot uses retained production data. The RR torus has a fixed relative correlator normalization, but its common string-amplitude constant and integrated amplitude remain undetermined.

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1 Conventions and scope

The authority for the formulas below is the local source snapshot, including local convention repairs, with base commit 488710430. Copied scientific files are unchanged, and their hashes are recorded in `provenance/source_origins.json`. The earlier machine notes are retained in `notes/originals`. We use

$$b = 1, \quad Q = 2, \quad c = \frac{3}{2} + 3Q^2 = \frac{27}{2}, \quad h_{\text{NS}}(P) = \frac{1 + P^2}{2}, \quad h_{\text{R}}(P) = \frac{9}{16} + \frac{P^2}{2}. \quad (1)$$

The ordinary central charge c , not $\hat{c} = 2c/3$, enters the recursions. Internal continuum momenta are positive. External momenta are analytically continued when necessary. A bar on a block denotes the specified antichiral frame at the same physical momenta; at complex momentum it is not generally complex conjugation of the whole numerical block.

Quantity	Block evaluation	Nonchiral convention
NS sphere	Multiprecision NS c -recursion and elliptic re-expansion	BRY G, H, J , with one odd-block sign conversion
R-containing sphere	Two Virasoro recursions, finite branching, assembled self-dual limit	Graded component sum, analytic dual, scattering half-sum fields
NS torus punctures	NS necklace c -recursion and conditioned R Ward/Gram contraction	Stored form phases and $E/2, O/2$
R torus punctures	NS/R radial trace and RR-pair OPE with handle traces	Original state sums followed by relative correlator prefactors

2 Explicit production three-point coefficients

Let G_B be the Barnes G -function. The exact $b = 1$ functions in the production structure-constant module are

$$\Upsilon_1(x) = G_B(x)G_B(2-x), \quad \Upsilon_1(1) = 1, \quad (2)$$

$$\begin{aligned} \Upsilon_{\text{NS}}(x) &= \Upsilon_1(x/2)\Upsilon_1(1+x/2) \\ &= \Gamma(x/2)\Gamma(1-x/2)G_B(x/2)^2G_B(1-x/2)^2, \end{aligned} \quad (3)$$

$$\Upsilon_{\text{R}}(x) = \Upsilon_1((x+1)/2)^2. \quad (4)$$

The code uses the second expression for Υ_{NS} , setting $\Upsilon_{\text{NS}}(0) = 0$ explicitly. Define

$$N_{\text{NS}}(P) = \frac{\Gamma(1+iP)}{\Gamma(1-iP)}\Upsilon_{\text{NS}}(2iP), \quad N_{\text{R}}(P) = \frac{\Gamma(\frac{1}{2}+iP)}{\Gamma(\frac{1}{2}-iP)}\Upsilon_{\text{R}}(2iP). \quad (5)$$

For three momenta write

$$S = P_1 + P_2 + P_3, \quad D_1 = P_2 + P_3 - P_1, \quad D_2 = P_1 + P_3 - P_2, \quad D_3 = P_1 + P_2 - P_3.$$

The actual NS functions are

$$C(P_1, P_2, P_3) = \frac{i}{2} \frac{\prod_j N_{\text{NS}}(P_j)}{\Upsilon_{\text{NS}}(1+iS) \prod_j \Upsilon_{\text{NS}}(1+iD_j)}, \quad (6)$$

$$\tilde{C}(P_1, P_2, P_3) = i \frac{\prod_j N_{\text{NS}}(P_j)}{\Upsilon_{\text{R}}(1+iS) \prod_j \Upsilon_{\text{R}}(1+iD_j)}. \quad (7)$$

They are real for positive real momenta in the BRY phase convention [1]. The displayed i 's are part of the analytic definitions. For $R(P_1)R(P_2)V(P_3)$, the two raw returned functions are

$$E(P_1, P_2; P_3) = -\frac{i}{2} \frac{N_R(P_1)N_R(P_2)N_{NS}(P_3)}{\Upsilon_R(1+iS)\Upsilon_R(1+iD_3)\Upsilon_{NS}(1+iD_1)\Upsilon_{NS}(1+iD_2)}, \quad (8)$$

$$O(P_1, P_2; P_3) = -\frac{i}{2} \frac{N_R(P_1)N_R(P_2)N_{NS}(P_3)}{\Upsilon_{NS}(1+iS)\Upsilon_{NS}(1+iD_3)\Upsilon_R(1+iD_1)\Upsilon_R(1+iD_2)}. \quad (9)$$

The first two arguments are Ramond and the third is NS.

2.1 Halves, phases and the factors of two and four

The sphere scattering fields and local torus trinion adapters use

$$c_+ = E/2, \quad c_- = O/2, \quad d_\epsilon = c_+ + \epsilon c_- = (E + \epsilon O)/2. \quad (10)$$

Here c_η labels an HJS sign form; d_ϵ labels a physical Ramond-family ground three-point amplitude. The NS graded sewing coefficients are

$$\Gamma_0 = C, \quad \Gamma_1 = i\tilde{C}, \quad -\Gamma_{L,1}\Gamma_{R,1} = +\tilde{C}_L\tilde{C}_R. \quad (11)$$

The same i occurs at both vertices in a bilinear coefficient product; using an absolute square would lose this sign.

The independent sphere sewing-metric audit finds that the half-sum scattering field has effective

$$D_{R,sc} = \frac{1}{2}D_{NS}, \quad D_{NS}(P, P') = \pi\delta(P - P'). \quad (12)$$

For an NS identity insertion $P_3 = i(1 - \varepsilon)$, $(E + O)/(2C) \rightarrow 1/2$. The R-exchange density absorbs the relative inverse metric: a printed common dP/π alone does not determine the metric. The finite reduced R ground vector still has unit norm.

The other machine note's equal-metric dictionary is $\hat{c} = (E, O)$. A consistent conversion is

$$\hat{R} = \sqrt{2}R_{sc}, \quad \hat{D}_R = 2D_{R,sc}, \quad \hat{d}_\epsilon = 2d_\epsilon. \quad (13)$$

It multiplies a mixed Liouville correlator by two and a four-R Liouville correlator by four. The coefficients of the same physical axion vertex then contribute $1/\sqrt{2}$ per R leg, canceling this change in the amplitude. Doubling two vertices with all other data fixed instead multiplies their contribution by four. No extra multiplier of this kind is applied here. The later RR torus factors in Section 5.2 are an overall correlator normalization and do not change state metrics.

3 Sphere four-point decompositions

Momenta (P_1, P_2, P_3, P_4) are assigned to $(0, z, 1, \infty)$. Evaluated blocks include coordinate powers, but exclude structure constants and spectral measures.

3.1 Four NS external states

Let P_e, P_o be the BRY-facing bottom-field blocks and D_e, D_o the blocks with $G_{-1/2}$ on legs 2 and 3. In terms of the stored fixed-parity blocks,

$$P_e = F_e, \quad P_o = -F_o, \quad D_e = F_e^{**}, \quad D_o = -F_o^{**}.$$

The stored unstarred odd block begins $F_o = -z^{h-h_1-h_2+1/2}/(2h) + \dots$. Put $C_L = C(P_4, P_3, P)$, $C_R = C(P, P_2, P_1)$, and similarly for $\tilde{C}$. The implemented decomposition is

$$G = \int_0^\infty \frac{dP}{\pi} [C_L C_R P_e \bar{P}_e + \tilde{C}_L \tilde{C}_R P_o \bar{P}_o], \quad (14)$$

$$H = - \int_0^\infty \frac{dP}{\pi} [\tilde{C}_L \tilde{C}_R D_e \bar{D}_e + C_L C_R D_o \bar{D}_o], \quad (15)$$

$$J = - \int_0^\infty \frac{dP}{\pi} \left[C_L C_R \left(\frac{D_o \bar{P}_e}{1 - \bar{z}} + \frac{P_e \bar{D}_o}{1 - z} \right) + \tilde{C}_L \tilde{C}_R \left(\frac{D_e \bar{P}_o}{1 - \bar{z}} + \frac{P_o \bar{D}_e}{1 - z} \right) \right]. \quad (16)$$

The antichiral functions evaluate the same analytic coefficient table at $\bar{z}$. H, J include the specific descendant and free-fermion combinations of BRY Appendix A; they are not arbitrary scalar four-point functions.

3.2 R external fields: explicit component sum

The stored families and ordered external sectors are

family	(s_1, s_2, s_3, s_4)	internal sector
<code>mixed_ns</code>	(R, R, NS, NS)	NS
<code>mixed_r</code>	(NS, R, R, NS)	R
<code>rrrr</code>	(R, R, R, R)	NS.

An NS component is the pair $(a, \bar{a})$ in $G_{-1/2}^a \bar{G}_{-1/2}^{\bar{a}} V$, in that fixed order. An R family $e = 0, 1$ has coefficients

$$u_0 = \frac{1}{\sqrt{2}}(1, 0, 0, -i)^T, \quad u_1 = \frac{1}{\sqrt{2}}(0, 1, 1, 0)^T \quad (17)$$

in the ground basis $(++, +-, -+, --)$. For an NS endpoint u is instead the delta function selecting its component bits. Middle R fields have chiral block bit zero; their physical family enters

$$V_e^\eta(r, \bar{r}) = \delta_{r+\bar{r}, e \bmod 2} \frac{1}{\sqrt{2}} \begin{cases} (-i)^r, & e = 0, \\ \eta, & e = 1. \end{cases} \quad (18)$$

The stripped spectral weights have no implicit fitted constant. Write $\alpha_j = 1 + iP_j$, $\alpha_\pm = 1 \pm iP$, and define

$$\mathcal{D}_t(a, b, c) = \Upsilon_{S_t}(a + b + c - 2) \Upsilon_{S_t}(a + b - c) \Upsilon_{S_t}(a + c - b) \Upsilon_{S_t}(b + c - a), \quad S_0 = \text{NS}, S_1 = \text{R}, \quad (19)$$

$$\mathcal{R}_\eta(n; r, s) = \Upsilon_{A_\eta}(n + r + s - 2) \Upsilon_{A_\eta}(r + s - n) \Upsilon_{B_\eta}(n + r - s) \Upsilon_{B_\eta}(n + s - r), \quad (20)$$

$$(A_+, B_+) = (\text{R}, \text{NS}), \quad (A_-, B_-) = (\text{NS}, \text{R}), \quad W_S(P) = \Upsilon_S(2\alpha_+) \Upsilon_S(2\alpha_-). \quad (21)$$

The densities used by the code are

$$\rho_{\text{mixed_ns}}(t; 1, \eta_R) = \frac{2^t W_{\text{NS}}(P)}{\mathcal{D}_t(\alpha_4, \alpha_3, \alpha_+) \mathcal{R}_{\eta_R}(\alpha_-; \alpha_2, \alpha_1)}, \quad (22)$$

$$\rho_{\text{mixed_r}}(0; \eta_L, \eta_R) = \frac{W_{\text{R}}(P)}{\mathcal{R}_{\eta_L}(\alpha_4; \alpha_3, \alpha_+) \mathcal{R}_{\eta_R}(\alpha_1; \alpha_2, \alpha_-)}, \quad (23)$$

$$\rho_{\text{rrrr}}(0; \eta_L, \eta_R) = \frac{W_{\text{NS}}(P)}{\mathcal{R}_{\eta_L}(\alpha_+; \alpha_3, \alpha_4) \mathcal{R}_{\eta_R}(\alpha_-; \alpha_2, \alpha_1)}. \quad (24)$$

The external factor restoring these stripped densities is

$$\mathcal{N} = \frac{(-1)^{n_R/2}}{2^{2+n_R/2}} \prod_{j=1}^4 N_{s_j}(P_j) = \begin{cases} -\frac{1}{8} \prod_j N_{s_j}(P_j), & n_R = 2, \\ \frac{1}{16} \prod_j N_R(P_j), & n_R = 4. \end{cases} \quad (25)$$

Sum independently over $k, \bar{k} = 0, 1$, the endpoint components $(a_1, \bar{a}_1), (a_4, \bar{a}_4)$, and $\eta_L, \eta_R = \pm 1$. Set $a_2 = \bar{a}_2 = 0$. On leg 3 use its NS bits, or zero for R. All following bit equations are modulo two:

$$r = k + a_1, \quad \bar{r} = \bar{k} + \bar{a}_1, \quad l = a_4 + k, \quad \bar{l} = \bar{a}_4 + \bar{k}.$$

For mixed NS, restrict $\eta_L = 1$, set $t = l + a_3$, $\bar{t} = \bar{l} + \bar{a}_3$, and take

$$\mathcal{L} = \delta_{t, \bar{t}} (-1)^{t(1+\bar{a}_3)} i^t, \quad \rho = \rho_{\text{mixed_ns}}(t; 1, \eta_R).$$

For the other families take $\mathcal{L} = V_{e_3}^{\eta_L}(l, \bar{l})$ and their $t = 0$ density. Let $\chi_R = 1$ only for internal R, and $\eta'_R = (-1)^{\chi_R} \eta_R$. The full correlator is

$$\begin{aligned} \mathcal{G}_e(z, \bar{z}) = \int_0^\infty \frac{dP}{\pi} \mathcal{N} \sum u_{e_1}(a_1, \bar{a}_1) u_{e_4}(a_4, \bar{a}_4)^* (-1)^{\bar{r}a_1 + \bar{l}k + \chi_R(r+\bar{r})} \mathcal{L} V_{e_2}^{\eta_R}(r, \bar{r}) \rho \\ \times \mathcal{F}_k^{\eta_L, \eta'_R}(z; \mathbf{a}) \bar{\mathcal{F}}_{\bar{k}}^{\eta_L, \eta'_R}(\bar{z}; \bar{\mathbf{a}}). \end{aligned} \quad (26)$$

The star on u_{e_4} conjugates its fixed phase, not P_4 . Both the internal-R sign reflection and the extra parity factor are essential. The production does not replace this sum by a guessed $\sum |\mathcal{F}|^2$. For complex momenta it constructs a dual table at $(\mathbf{P}^*, P^*)$, then conjugates its value: $\bar{\mathcal{F}}(\bar{z}; \mathbf{P}) = [\mathcal{F}(z; \mathbf{P}^*)]^*$. The native sign and small-representation conventions are those of the corrected Ramond formulas in [2, 3].

4 How the sphere blocks are evaluated

4.1 Definition and NS coefficient recursion

The local SCA has

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}, \quad (27)$$

$$[L_m, G_r] = (m/2 - r)G_{m+r}, \quad \{G_r, G_s\} = 2L_{r+s} + \frac{c}{3}(r^2 - \frac{1}{4})\delta_{r+s,0}. \quad (28)$$

PBW bases put ordered negative L modes before distinct ordered negative G modes, reducing $G_{-r}^2 = L_{-2r}$. R zero modes act on the ground doublet. At level ℓ the definition is the ordered Ward contraction

$$f_\ell = \sum_{|I|=|J|=\ell} \rho_L(v_4, v_3, I) (B_\ell^{-1})^{IJ} \rho_R(J, v_2, v_1). \quad (29)$$

Ground-frame factors belong to ρ and B , once. The NS seeds are $\rho_0(\nu, \nu, \nu) = 1$ and $\rho_1(\nu, G_{-1/2}\nu, \nu) = 1$.

The NS code uses a memoized c -recursion to compute (29). Write $F_N = f_{N/2}$, let s_2, s_3 be the external star bits, and set $A = h + h_3 - h_4$, $B = h + h_2 - h_1$. Its large- c seeds are

$$f_{2m}^{(\infty)} = \frac{(A + s_3/2)_m (B + s_2/2)_m}{m! (2h)_m}, \quad (30)$$

$$f_{2m+1}^{(\infty)} = -\frac{X_{s_3}(A, m) X_{s_2}(B, m)}{m! (2h)_{m+1}}, \quad X_0(x, m) = (x + \frac{1}{2})_m, \quad X_1(x, m) = (x)_{m+1}. \quad (31)$$

For $r \geq 2, s \geq 1, r+s$ even, define

$$\Delta = \sqrt{16h^2 + 8(rs-1)h + (r-s)^2}, \quad x_{rs} = -\frac{4h + rs - 1 + \Delta}{r^2 - 1}, \quad (32)$$

$$b_{rs} = \sqrt{x_{rs}}, \quad c_{rs} = \frac{15}{2} + 3x_{rs} + 3/x_{rs}, \quad J_{rs} = -\partial_h c_{rs}. \quad (33)$$

The code chooses the sign of Δ giving the larger-magnitude numerator, the standard continuation of positive real h . Let

$$A_{rs}(b) = \frac{1}{2} \prod_{\substack{p=1-r, \dots, r; \\ p+q \text{ even} \\ (p,q) \neq (0,0), (r,s)}} \frac{\sqrt{2}}{pb + q/b}, \quad (34)$$

$$\mathcal{P}_{rs}^a(h_i, h_j; b) = \prod_{\substack{p=1-r, 3-r, \dots, r-1 \\ q=1-s, 3-s, \dots, s-1 \\ p+q-r-s \equiv 2(1-a) \pmod{4}}} \frac{(\lambda_i - \lambda_j + pb + q/b)(\lambda_i + \lambda_j + pb + q/b)}{8},$$

$$\lambda_j = \sqrt{(b + b^{-1})^2 - 8h_j}. \quad (35)$$

Then

$$F_N(h, c) = f_N^{(\infty)}(h) + \sum_{\substack{r \geq 2, s \geq 1, r+s \text{ even} \\ rs \leq N}} \frac{R_{rs}^{(N)}(h)}{c - c_{rs}(h)} F_{N-rs}(h + rs/2, c_{rs}), \quad (36)$$

$$R_{rs}^{(N)}(h) = (-1)^{rs} J_{rs} A_{rs} \mathcal{P}_{rs}^{s_2 \oplus (N \bmod 2)}(h_1, h_2) \mathcal{P}_{rs}^{s_3 \oplus (N \bmod 2)}(h_4, h_3).$$

The endpoint $rs = N$ is included. Each call lowers N ; the supplied sphere examples use 60-digit coefficient arithmetic.

The local block is $z^{h-h_1-h_2-s_2/2} \sum_N F_N z^{N/2}$ with one parity retained. Put $q = e^{i\pi\tau_z}$, $\tau_z = iK(1-z)/K(z)$ and $d_j = h_j + s_j/2$, with $s_1 = s_4 = 0$. Its elliptic prefactor is

$$(16q)^{h-Q^2/8} z^{Q^2/8-d_1-d_2} (1-z)^{Q^2/8-d_2-d_3} \theta_3(q)^{3Q^2/2-4} \sum_j d_j. \quad (37)$$

The remaining coefficients are obtained by the formal substitution $z = \theta_2(q)^4/\theta_3(q)^4$ and division by this prefactor. No new fitted elliptic seed is inserted. The ordinary order counts coefficients in each parity. BRY order L instead retains even powers through q^L and odd powers through $q^{L-1/2}$.

4.2 R blocks: branching, auxiliary division and self-dual limit

The production tensors the SCA with an auxiliary Majorana module and decomposes it into two commuting Virasoro modules at generic b . The correlated roots are

$$b_1 = \sqrt{\frac{2b^2}{1-b^2}}, \quad b_2^{-1} = \sqrt{\frac{2}{b^2-1}}, \quad d_1 = \sqrt{2-2b^2}, \quad d_2 = -b_2^{-1}d_1/b_1. \quad (38)$$

For sphere momentum P , the branch n has

$$p_n^{(1)} = -iP/d_1 + nb_1, \quad p_n^{(2)} = -iP/d_2 + n/b_2, \quad (39)$$

$$c_j = 1 + 6(b_j + b_j^{-1})^2, \quad h_n^{(j)} = \frac{1}{4}(b_j + b_j^{-1})^2 - (p_n^{(j)})^2. \quad (40)$$

The sign $-iP$ implements the sphere frame $\beta = -iP/\sqrt{2}$. The marked torus Ward source has its separately fixed $\beta = +iP/\sqrt{2}$, and its tensors are not silently substituted here.

The finite tensor PBW basis at the branch grade is constructed. The known $L_0^{(1)}$ eigenvalue $h_n^{(1)}$ gives a constrained linear solve for the highest vector v_n , with bilinear norm $v_n^T B v_n$. Diagonal balancing, isolation tests and high-precision residual checks precede the branch sum; assembly retains 60 digits. An external state is resolved by a finite linear solve in branch vectors, with labels 0 for an NS bottom state, $\pm 1/2$ for its $G_{-1/2}$ component, and $\pm 1/4$ for an R ground.

In the linear sign frame the finite enlarged descendant block is

$$\mathcal{F}_{\text{enl}}(z) = \sum_{n,\nu} \frac{\prod_j t_{j,\nu_j} \rho_L(v_{\nu_4}, v_{\nu_3}, v_n) \rho_R(v_n, v_{\nu_2}, v_{\nu_1})}{\langle v_n, v_n \rangle} z^{g_n} \mathcal{V}_1(z) \mathcal{V}_2(z). \quad (41)$$

Here t are external resolution coefficients, $g_n = 2n^2$ in NS and $g_n = 2n^2 - 1/8$ in R. Only $2g_n \leq N$ is kept; NS also requires $2n \bmod 2 = k$. Each $\mathcal{V}_j$ is the unit-leading ordinary Virasoro block at the displayed branch weights, computed by sphere c -recursion; their series are convolved through the remaining order. Explicitly, its coefficients are

$$v_m(c, h) = \frac{(h + h_2 - h_1)_m (h + h_3 - h_4)_m}{m! (2h)_m} + \sum_{\substack{r \geq 2, s \geq 1 \\ rs \leq m}} \frac{J_{rs}^V A_{rs}^V P_{rs}^V(h_4, h_3) P_{rs}^V(h_1, h_2)}{c - c_{rs}^V} v_{m-rs}(c_{rs}^V, h + rs), \quad (42)$$

$$x_{rs}^V = \frac{rs - 1 + 2h + \sqrt{(r-s)^2 + 4(rs-1)h + 4h^2}}{1 - r^2}, \quad c_{rs}^V = 13 + 6(x_{rs}^V + 1/x_{rs}^V), \quad (43)$$

$$J_{rs}^V = -\partial_h c_{rs}^V, \quad A_{rs}^V = \frac{1}{2} \prod_{\substack{p=1-r, \dots, r; \\ (p,q) \neq (0,0), (r,s)}} \quad q=1-s, \dots, s \quad (pb + q/b)^{-1}, \quad (44)$$

$$P_{rs}^V(h_i, h_j) = \prod_{\substack{p=1-r, 3-r, \dots, r-1 \\ q=1-s, 3-s, \dots, s-1}} \frac{(\lambda_i + \lambda_j + pb + q/b)(\lambda_i - \lambda_j + pb + q/b)}{4}. \quad (45)$$

Here $b = \sqrt{x_{rs}^V}$ and $\lambda_j = \sqrt{(b + b^{-1})^2 - 4h_j}$ only within the Virasoro residue. If the displayed root is numerically zero, the code uses the other finite root. It algebraically cancels common factors in $J^V A^V$ before numerical evaluation; unresolved poles raise an error. This recurrence is evaluated at 60 digits in the production precision bootstrap. The ordinary Virasoro recurrence follows the sphere specialization of [4]; the finite branching and frame conversion are specified by the preserved local sources.

Branch vertices are evaluated by expanding the finite vectors and applying the ordered physical and auxiliary Ward forms with their graded tensor signs. No unverified closed Ramond branching product is used as the definition.

Remove the external polynomial-ground scales and divide the formal series, coefficient by coefficient, by the unit-leading auxiliary block

$$A_{\text{mixed_ns}} = 1, \quad A_{\text{mixed_r}} = (1 - z)^{-1/8}, \quad A_{\text{rrrr}} = (1 - z)^{-1/8} \sqrt{\frac{1 + \sqrt{1 - z}}{2}}. \quad (46)$$

Their leading z powers are already stripped. Define

$$T = \frac{1}{2} \begin{pmatrix} 1 - i & 1 + i \\ 1 + i & 1 - i \end{pmatrix}, \quad T_f = \text{diag}(1, (-1)^f) T.$$

The native sign vector is $M^{-1}\mathcal{F}_{\text{linear}}$, with $M = I$ for mixed NS, $I_2 \otimes T_{k+a_2+a_1}$ for mixed R, and $i^{a_4}T_{a_4+a_3+k} \otimes I_2$ for four R, in sign order $+, -$.

At fixed momenta and component labels, the *assembled native coefficient* is extrapolated as

$$f(1) \simeq \frac{4(1+\epsilon)f(1+\epsilon) - (1+2\epsilon)f(1+2\epsilon)}{3+2\epsilon}, \quad \epsilon = 0.01. \quad (47)$$

This cancels the leading dependence on $Q-2 = (b-1)^2/b$. Formal $O(\epsilon^4)$ accuracy assumes analyticity and b -duality; it is not a certified error estimate. Singular individual Virasoro factors are not extrapolated separately.

Finally use the exact $b = 1$ prefactors. For $d_j = h_{s_j}(P_j) + a_j/2$ on NS legs and $d_j = h_R(P_j)$ on R, and $B_0 = Q^2/8 = 1/2$,

$$\mathcal{F}(z) = (16q)^{P^2/2} z^A (1-z)^B \theta_3(q)^K \sum_{j=0}^N h_j t^j, \quad t = e^{i\pi\tau z/2}, \quad (48)$$

$$A = B_0 - d_1 - d_2 + \frac{1}{16} \mathbf{1}_{\text{mixed_r}}, \quad (49)$$

$$B = B_0 - d_2 - d_3 + \frac{1}{16} \mathbf{1}_{\text{mixed_ns}}, \quad K = 12B_0 - 4 \sum_j d_j + \frac{1}{2} \mathbf{1}_{\text{not rrrr}}. \quad (50)$$

The h_j follow by finite formal substitution of the calculated z -coefficients. The half-nome is taken from τ_z , not an independently chosen square root. Cuts need a consistent marked lift.

5 Torus two-point decompositions

5.1 NS external fields: production necklace

The punctures are $0, z$, with periods $1, \tau$, and

$$\log q_0 = 2\pi iz, \quad \log q_1 = 2\pi i(\tau - z), \quad 0 < \text{Im } z < \text{Im } \tau. \quad (51)$$

Keep these unwrapped logs. Both external Liouville momenta are ω ; internal P_0, P_1 are independent. For homogeneous forms $f = (f_0, f_1)$ and external word w , the descendant series is

$$F_{w,f} = \sum_{\ell_0, \ell_1} q_0^{\ell_0} q_1^{\ell_1} \sum \rho_{f_0}(I_1, w_0, J_0) (B_0^{-1})^{J_0 I_0} \rho_{f_1}(I_0, w_1, J_1) (B_1^{-1})^{J_1 I_1}. \quad (52)$$

The source vertex contains its ordered phases. In NS, $N_j = 2\ell_j$ and $N_{v-1} + N_v + w_v = f_v \pmod{2}$, with $w_v = 0, 1$ for bottom and $G_{-1/2}$ fields.

The NS graph recursion sums over a Kac pole on either edge, subtracts rs from that edge's twice-level, shifts its weight by $rs/2$, and flips both incident form bits if rs is odd. The kernel is $(-1)^{rs} J A \mathcal{P}_L P_R$, using the functions in (36); for edge e the endpoint weight pairs are (h_{e+1}, d_{e+1}) and (h_{e-1}, d_e) , cyclically. The regular term is the global $osp(1|2)$ necklace convolved with the non-global vacuum modes, which shift both edge levels equally. Explicitly, with $\mathbf{q} = q_0 q_1$ and $\sigma = (-1)^{f_0+f_1}$, that vacuum factor is

$$\chi_{\text{ng}}(\mathbf{q}) = \prod_{n=2}^{\infty} \frac{1 + \sigma \mathbf{q}^{n-1/2}}{1 - \mathbf{q}^n} = \sum_{K \geq 0} \chi_K \mathbf{q}^{K/2}, \quad f_{N_0, N_1}^{(\infty)} = \sum_{K=0}^{\min(N_0, N_1)} \chi_K f_{N_0-K, N_1-K}^{\text{global}}.$$

Global states $L_{-1}^n G_{-1/2}^\epsilon |h\rangle$ have norm $n!(2h)_{n+\epsilon}$. Their forms follow from the finite global Ward identities. The adapter multiplies the final coefficient by $(-1)^{\sum_v f_v N_v}$ to obtain the production

component tensor. At total twice-level 15 and above, preparation tests successive 60–120 digit evaluations to scaled tolerance 10^{-11} . Guarded confluent cases use the source’s checked contour average.

R internal edges use a direct conditioned PBW calculation instead. At each level both ground parities are kept, Gram matrices are split by full state parity, diagonally equilibrated and Cholesky factored. Whitening each vertex and taking the finite matrix trace evaluates (52) without explicitly forming a poorly conditioned inverse. External NS-weight dependence is evaluated as a finite polynomial at the actual complex energy. An R supertrace inserts full state parity, including its ground parity; integer descendant level is not that parity. The cutoff is total twice-level $N_0 + N_1 \leq N_{\max}$.

The adapter returns GG/GG, GG/PP, PP/GG and PP/PP cylinder components. Let $H_G = -(-1)^f F_G$, $H_P = F_P$, with $f = f_0 = f_1$. The PP pairing phase is one for NS internal states and $(-1)^f$ for R internal states. The fixed-momentum components are

$$\mathcal{L}_\delta^{ab} = \frac{1}{\pi^2} \prod_{j=0}^1 e^{(h_s(P_j) - c/24)(\log q_j + \overline{\log q_j})} \sum_{\lambda} K_{\lambda} H_{a,\lambda} \bar{H}_{b,\lambda}, \quad a, b = G, P, \quad (53)$$

where

$$K_f^{\text{NS}} = \prod_{v=0}^1 C_{f_v}(P_{v-1}, \omega, P_v), \quad (C_0, C_1) = (C, \tilde{C}), \quad (54)$$

$$K_{f,\eta_0,\eta_1}^{\text{R}} = (-1)^f \prod_{v=0}^1 \left[\frac{B_{\eta_v}(P_{v-1}, P_v; \omega)}{2} \right], \quad (B_+, B_-) = (E, O). \quad (55)$$

In particular, two Ramond trinions supply $1/4$, with the odd pairing sign retained. More generally, the code’s ordered pairing phase for external words a, b is

$$\Phi_s(f; a, b) = (-i)^{\sum b_v + \mathbf{1}_{s=\text{NS}}} \sum f_v (-1)^{\sum f_v b_v + \sum_{i < j} (f_i + b_i)(f_j + a_j)}.$$

The component adapter uses the PP weights $a = b = (0, 0)$, after converting the GG chiral series as above. NS supertrace supplies $(-1)^{N_1}$ in each chiral series. R supertrace uses the state-parity insertion described above. The antichiral table keeps the same analytic coefficients and conjugates the geometric powers. External 2π coordinate factors, the GG factors $\pm 2\pi i$ in the PCO terms, and time/ghost factors are outside (53). The adapter states that convention. Its R supertrace is a Liouville component, not a completed odd-spin string density.

5.2 R external fields: radial and OPE channels

An R puncture changes the propagating sector. The radial sum has one NS and one R edge, with times

$$(t_{\text{NS}}, t_{\text{R}}) = \begin{cases} (z, \tau - z), & \delta = 1, 2, \\ (\tau - z, z), & \delta = 3, 4, \end{cases}$$

where δ labels $\theta_1, \theta_2, \theta_3, \theta_4$. Let $A_{n,\bar{n},r,\bar{r}}^{e,\delta}$ be the two full physical NSRR vertices contracted with inverse NS and reduced-R Grams at levels $(n/2, \bar{n}/2; r, \bar{r})$, computed by the numeric NRR Ward engine with (17) and $c_\eta = B_\eta/2$. In explicit matrix indices this means

$$A^{e,\delta} = \sum_{i,j,k,l} M_{ij}^e s_j^\delta (B_{\text{NS}}^{-1})_{ik} (B_{\text{R,red}}^{-1})_{jl} (M^{e,\sharp})_{kl}, \quad M_{ij}^e = \sum_{\eta=\pm} c_\eta \rho_{ij}^{e,\eta}. \quad (56)$$

The NS indices i, k include both chiralities. The R indices j, l label $(w, \bar{w}, e_R)$ in the reduced physical basis. Here $s_j^1 = (-1)^{|w|+|\bar{w}|+e_R}$, and $s_j^\delta = 1$ otherwise. The sharp is the adjoint of the vertex evaluated

at ω^* and conjugated couplings, so that the completed result is analytic in ω . The reduced R Gram is computed by embedding with (17) into the tensor product of full chiral Hermitian Grams and then inverting it.

For clarity, the marked NRR chiral engine uses

$$G_0|a\rangle = \beta e^{i\pi/4} i^a |1-a\rangle, \quad \beta = iP/\sqrt{2},$$

and the ground form at (NS, R, R) has entries $(00, 01, 10, 11) = (1, 1, i\eta, \eta)$, restricted by $p_\phi + a_2 + a_3 = f \pmod{2}$. Its positive-mode actions and Gram entries follow by commuting the displayed SCA through ordered PBW words; the generalized Ramond Ward recursion is preserved in `numeric_nrr.py`. The sphere branch computation instead uses $G_0|a\rangle = (P/\sqrt{2})e^{-(-1)^a i\pi/4} |1-a\rangle$ in its own frame. These formulae explain why the frames are kept distinct. The radial expansion is

$$L_{\delta, \text{raw}}^{ee} = (2\pi)^{4h_R(\omega)} \sum_{n, \bar{n}, r, \bar{r}} A_{n, \bar{n}, r, \bar{r}}^{e, \delta} \exp \left\{ 2\pi i \left[t_{\text{NS}}(h_{\text{NS}} + n/2 - c/24) - \bar{t}_{\text{NS}}(h_{\text{NS}} + \bar{n}/2 - c/24) \right. \right. \\ \left. \left. + t_{\text{R}}(h_{\text{R}} + r - c/24) - \bar{t}_{\text{R}}(h_{\text{R}} + \bar{r} - c/24) \right] \right\}. \quad (57)$$

For $\delta = 4$ include $(-1)^{n+\bar{n}}$; for $\delta = 1$ use the full R-state supertrace in A . The returned $e = 0, 1$ components are separate. The same Ward engine gives descendant insertions, including the checked relation

$$L_{G_0, \bar{G}_0}^{++} = -i\omega^2 L^{++}/2, \quad L_{G_0, \bar{G}_0}^{--} = +i\omega^2 L^{--}/2.$$

The independent OPE joins the two R punctures onto an NS bridge of momentum P , then traces that descendant around an NS/R handle of momentum p . Both bridge families $f = 0, 1$ are kept. For mode $m = 0, 1$, its chiral block is

$$\mathcal{B}_m^f = (2\pi i)^{h_{\text{NS}}(P) + f/2} z^{h_{\text{NS}}(P) + f/2 - h_{R,z} - h_{R,0} - m} \sum_{j=0}^J \sum_{\ell \leq L} b_{j,\ell}^{f,m} z^j e^{2\pi i \tau (h_{\text{handle}} + \ell - c/24)}. \quad (58)$$

For no insertion use $m = 0$ in the exponent and a bare Ward vector. Each coefficient contracts the RR-pair Ward vector, inverse NS bridge Gram and handle one-point Ward trace. The plane coordinate $v = e^{2\pi i z} - 1$ is converted by expanding

$$e^{2\pi i z h_{R,z}} \left(\frac{e^{2\pi i z} - 1}{2\pi i z} \right)^{h_{\text{NS}} + f/2 - h_{R,z} - h_{R,0} - m + j}.$$

For G_{-1} there is also the source's $\frac{3}{4}(2\pi i)G_0$ connection term. Bridge order, NS handle twice-level and R handle integer level are independent cutoffs. The implemented contractible patch has $0 < |z| < 1/4$ and the additional local geometry checks of the source.

The RR-pair coefficient is $c_\eta(\omega, \omega; P)$. The handle coefficients are

$$H_{\delta, f, \eta} = \begin{cases} i^f C_f(p, P, p) \delta_{\eta, +}, & \delta = 3, 4, \\ i^f c_\eta(p, p; P)/2, & \delta = 1, 2. \end{cases} \quad (59)$$

The last $1/2$ removes the doubled Clifford multiplicity in the product of full chiral traces. In components the precise assembly is

$$L_{\delta, \text{OPE}, \text{raw}}^{ee} = \sum_{\eta, \eta_h, f} c_\eta H_{\delta, f, \eta_h} \sum_{i, j=1}^4 S_{ij}^{e, f} \mathcal{B}_{i, \eta, \eta_h}^f \bar{\mathcal{B}}_{j, \eta, \eta_h}^f. \quad (60)$$

For $i = (z_h, o_h), j = (z_a, o_a)$, and $u = 0$ bare or $u = 1$ for one odd mode in each chirality, the explicit matrix is

$$S_{ij}^{e,f} = u_e(z_h, z_a) u_e(o_h, o_a) (-1)^{u o_h + f(f + z_h + o_h + u) + z_a(o_h + u) + f}.$$

The anti OPE engine receives $-\omega^*$, so that $\beta = i\omega/\sqrt{2}$ is conjugated, and then its table is conjugated. Using $+\omega^*$ gives the same weights but the wrong zero-mode phases.

The current local wrappers apply, after each original state sum,

$$L_{\delta, \text{radial}} = 2L_{\delta, \text{radial, raw}}, \quad L_{\delta, \text{OPE}} = (2, 2, 1, 1)_{\delta} L_{\delta, \text{OPE, raw}}. \quad (61)$$

These constant correlator prefactors are relative to the OPE NS-handle reference. They do not change Grams, primary norms, trinions or chiral coefficients. The original Clifford 1/2 is kept. The package calls the existing normalized wrapper classes, so these prefactors are applied once. Radial and OPE channels use different internal momentum pairs: their fixed-momentum integrands need not agree. Compare their respective spectral integrals. The common string-amplitude normalization remains `None`.

6 Cross-channel checks and amplitude plot

6.1 Fresh checks in this package

The file `validation/science.json` records the following checks. All 576 fresh coefficients for R-containing sphere functions, through twice-level two in all three families and both analytic chiralities, match their retained order-five production coefficients exactly at stored precision. This is a fresh low-level computation, not a fresh order-five production run.

NS primary crossing was rerun at $c = 27/2$, BRV order eight, 36 momentum nodes on $0 < P < 5$, and $(P_1, P_2, P_3, P_4) = (1/2, 1/3, 1/4, 3/5)$. The crossing is $z \mapsto 1 - z$, $P_1 \leftrightarrow P_3$:

z	G_s	G_t	Relative discrepancy
0.37	0.221609720986587	0.221609720986640	$2.39 \cdot 10^{-13}$
$0.37 + 0.11i$	0.211719203999165	0.211719203999662	$2.35 \cdot 10^{-12}$

The earlier retained G, H, J audit also tests inversion of H, J : its order-eight discrepancies at the complex point are 2.34×10^{-5} , 1.96×10^{-5} . Those descendant spectral checks were not rerun here.

Twelve NS torus components at $\omega = 0.2 + i/4$ and total twice-level two reproduce the original production even-spin assembler after restoring external free factors, with scaled error 3.84×10^{-20} . RR torus prefactor, G_0 , and global level-one checks have maximum residual 1.12×10^{-16} ; the chiral blocks change by exactly zero under the overall prefactor.

The command `run.py cross-check` freshly assembles the full sphere production kernels and spectral sums from the included 96 order-five banks at $\omega = i/4$, verifying their checksums. No moduli integral is performed. The u -chart comparison includes the complete-vertex Jacobian $|z|^{-4}$ or $|1 - z|^{-4}$. Individual chiral blocks are not asserted to be crossing invariant.

Comparison	Maximum relative discrepancy
Four R, s against t	$4.581 \cdot 10^{-5}$
Four R, s against u	$5.147 \cdot 10^{-5}$
Mixed, s against t	$1.104 \cdot 10^{-4}$
Mixed, s against u	$8.423 \cdot 10^{-5}$
Mixed picture placement, s	$1.460 \cdot 10^{-9}$
Mixed picture placement, t	$2.370 \cdot 10^{-9}$
Mixed picture placement, u	$2.393 \cdot 10^{-5}$

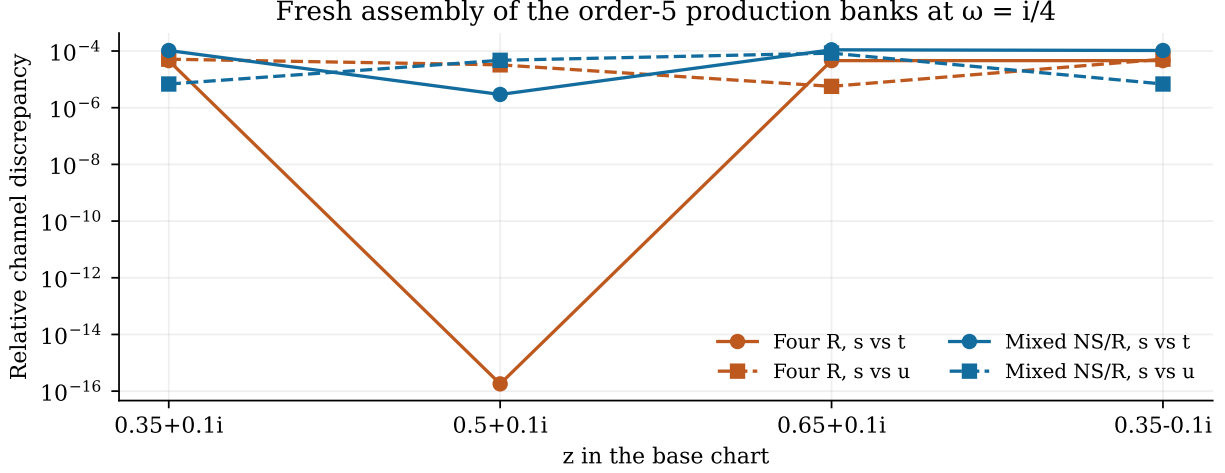


Figure 1: Fresh spectral assembly of retained order-five sphere production banks. Finite block order, momentum quadrature and self-dual extrapolation remain numerical approximations. Zero differences are plotted at the 10^{-16} display floor.

6.2 Retained RR torus evidence

The included normalization report records independent 1369-node and 676-node spectral integrations at $\tau = i$, $z = 0.04 + 0.1i$ and its modular image. With (61) and theta permutation (1, 4, 3, 2), the maximum relative modular error is below 1.10×10^{-6} ; the quadratures agree within 8.95×10^{-8} . The retained 64-node radial/OPE comparisons have maximum discrepancies 1.82×10^{-3} bare and 5.74×10^{-2} raised. Raised components still require block and spectral refinement. These larger integrations were not rerun while packaging; the report and its input hashes are supplied. They do not fix a common genus-one string-amplitude constant.

6.3 Type 0B sphere amplitude

Figure 2 is regenerated from the retained production scan without refitting or normalization adjustment. For the tachyon T and axion A , the incoming energy is $\omega = it$ and the three outgoing energies are $it/3$. The comparison target is

$$\mu_F^2 \mathcal{M}_{A \rightarrow ATT} = \mu_F^2 \mathcal{M}_{A \rightarrow AAA} = \frac{it^4(1-2t)}{27}, \quad (62)$$

with the energy delta function omitted. It is used only after worldsheet integration, not as an integrand coefficient. The scan used 32 spectral nodes, physical block orders four and five, $b = 1.01, 1.02$, 60-digit assembly, collision radius 0.18, and 24-by-48 bulk moduli quadrature. Maximum observed relative differences are about 0.02581% mixed and 0.00247% four R. These are comparison residuals, not certified absolute errors.

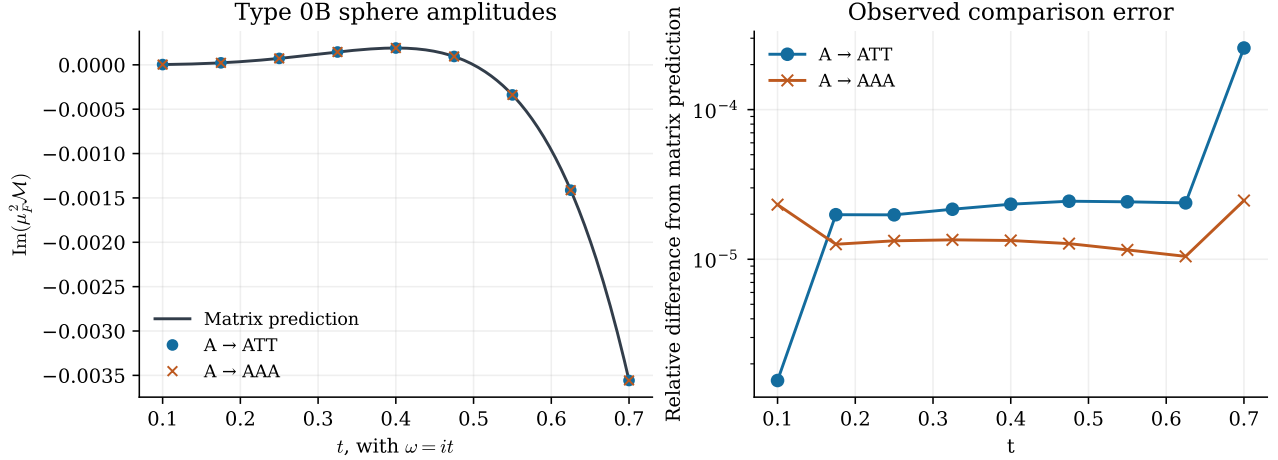


Figure 2: Type 0B genus-zero amplitudes in the local scattering dictionary. Points are retained worldsheet results; the line is (62). Right: measured relative difference. Data: notes/evidence/sphere_amplitude_scan.json. No torus amplitude is plotted: the current RR torus still lacks a fixed absolute string normalization and completed integrated result.

7 Reproduction

Use Python 3.11 or later from the standalone folder:

```
python3.11 -m venv .venv
.venv/bin/python -m pip install -r requirements-tested.txt
.venv/bin/python -I -B run.py verify
.venv/bin/python -I -B run.py examples --output-dir results
.venv/bin/python -I -B run.py test --output results/validation.json
.venv/bin/python -I -B run.py cross-check --output results/cross_channel.json
```

Edit the JSON examples to change momenta, insertion points or cutoffs. Complex numbers are [real,imaginary] pairs. The examples return blocks and fixed-momentum assemblies; the NS sphere also accepts a finite spectral quadrature. Every output states its included primary powers, frame factors and measures. The low example orders are execution checks, not convergence claims.

The included banks cover only the $\omega = i/4$ crossing check. The nine-energy amplitude scan is retained evidence for the plot, not a bundled complete nine-energy integration campaign. Regenerate both figures with tools/make_figures.py. Ramond zero momentum, confluent poles and branch-cut continuation need the corresponding limiting prescription.

The practical rules are to retain the local trinion dictionary, retain its metric and component contractions, apply the RR torus correlator prefactor only after the descendant sum, and include each spectral measure and external leg factor once.

References

- [1] B. Balthazar, V. A. Rodriguez and X. Yin, *The S-Matrix of 2D Type 0B String Theory, Part 1*, [arXiv:2201.05621](#), Sections 3.1–3.2 and Appendix A.
- [2] P. Suchanek, *Elliptic recursion for 4-point superconformal blocks and bootstrap in $N=1$ SLFT*, [arXiv:1012.2974](#).
- [3] L. Hadasz, Z. Jaskolski and P. Suchanek, *Elliptic recurrence representation of the $N=1$ superconformal blocks in the Ramond sector*, [arXiv:0810.1203](#).
- [4] M. Cho, S. Collier and X. Yin, *Recursive Representations of Arbitrary Virasoro Conformal Blocks*, [arXiv:1703.09805](#).
- [5] Local production source and hashed numerical evidence in this package. Source-specific phases and normalization qualifications describe these implementations and do not assert that every convention in the literature has been reconciled.